

Image Analysis Report

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Executive Summary

This inspection reveals high-severity localized corrosion concentrated along a horizontal weld seam. The presence of heavy scale and coating delamination suggests significant metal loss that could compromise the component's pressure-retaining or structural integrity.

Key Findings

#	Finding
1	Severe localized corrosion centered on a horizontal weld joint, indicating possible preferential weld attack.
2	Presence of thick, friable rust scale and exfoliation in the central area, suggesting an advanced corrosion state.
3	Extensive coating failure with delamination and under-film corrosion spreading from the primary site.
4	Likely deep pitting at the intersection of the weld and the base metal.
5	Widespread atmospheric corrosion spotting and rust staining on the surrounding substrate.
6	Actual remaining wall thickness cannot be determined from visual assessment and requires NDT validation.

Localized detections (VLM, approximate)

- Severe Localized Corrosion (conf. 95%): x=0.320, y=0.110, w=0.460, h=0.810
- Corroded Weld Seam (conf. 88%): x=0.000, y=0.450, w=1.000, h=0.120

Recommendations

#	Recommendation
1	Prioritize Ultrasonic Thickness (UT) testing to determine the percentage of wall loss in the pitted regions.
2	Abrasive blast clean to ISO 8501-1 Sa 2.5 (Near-White Metal) to fully expose the metal surface for detailed inspection.
3	Conduct Magnetic Particle Inspection (MPI) or Dye Penetrant Inspection (DPI) on the weld area to check for stress corrosion cracking or localized defects.
4	If wall thickness is below the engineering minimum (T-min), implement a code-compliant repair such as a weld overlay or structural patch.
5	Apply a high-performance three-coat epoxy/polyurethane system after surface preparation to prevent recurrence.
6	Review and potentially upgrade the external cathodic protection system if this is a buried or submerged asset.